



Synthetic Data for HTA and Real-World Evidence

Overcoming Data Access & Privacy Constraints to Enable Evidence Generation

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Purpose: A practical framework to outline synthetic data use in HTA and RWE: Distinguish operational support, evidence-planning and decision-relevant use cases for HTA and EU Joint Clinical Assessment (JCA).

WHY THIS MATTERS

- Synthetic data cannot compensate for limitations in the underlying evidence base. If gaps in the original data are insufficient, such as limited access to individual patient-level data, sparse or underpowered subgroups, privacy restrictions, population heterogeneity, or missingness, then these gaps will be reflected in the generated data.
- For an AMNOG context, this places clear requirements on how synthetic data can contribute: its evidentiary weight depends on the quality of the source data, the plausibility of model assumptions, and transparent validation against patient-relevant endpoints and the appropriate comparator

WHAT COUNTS AS SYNTHETIC DATA?

Operational definition: Data generated from a (closed form) model or a prediction from a black box model such as statistical parametric models, or machine learning algorithms. That process that generates artificial datasets is designed to replicate the certain properties of real data whilst preventing the disclosure of original sensitive patient-level individual information. Examples include Pharmacometrics generation based on biological mechanism and expert knowledge, parametric or agent-based simulation, Bayesian networks and generative adversarial networks.

Assumptions to examine preserved marginals and joint distributions, exchangeability, outcome mechanisms and fit-for-purpose validation.

HTA/RWE Use cases:

Synthetic patient records

Fully simulated individuals

Simulated trial populations

Scenario & power planning

Long-term Follow-Up

Outcome prediction

External control arm

Feasibility assessment and comparative studies

Synthetic RWE Registry

Analysis of restricted & sensitive data

Health Economic Model

Parameterization & model input

Key principle: As synthetic data moves closer to informing decision-making, the rigor applied to it must increase accordingly. Four factors become critical: validation of key assumptions, end-to-end traceability, external validation, and robust sensitivity analyses.

OPEN QUESTIONS – WE NEED YOUR PERSPECTIVE



Which use cases are acceptable as method-enabling versus evidence-informing in HTA or RWE?



Can synthetic data complement evidence for small subgroups and where are the limits? Minimum sample size for generation?



How should generation-process uncertainty be communicated alongside clinical uncertainty?



What about auditability? Can data be reproduced at request?



What validation standards, utility and quality metrics are needed before synthetic data can support decision-relevant analyses?



Which HTA agencies would be open to accepting evidence based on synthetic data?

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